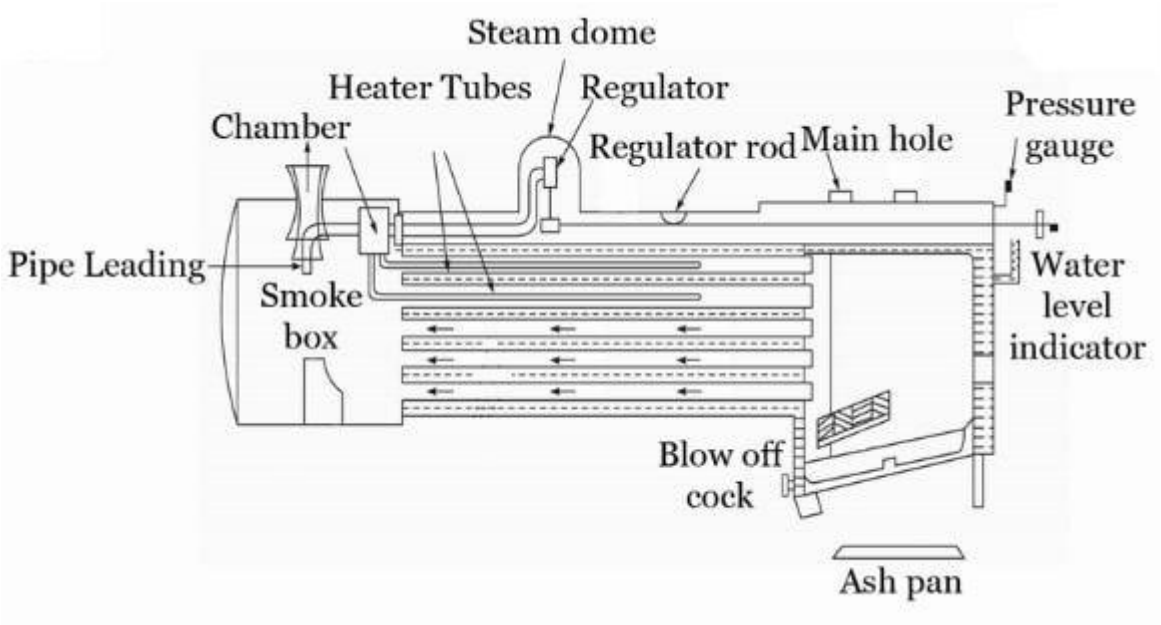
What is a Boiler?

A boiler is an enclosed vessel that provides a medium in the transfer of heat and pressure until it turns the water into steam. The heating surface of any boiler, like the hot gases, and liquid gets placed on either side. The boiler part that provides in making steam is the heating surface, expressed in square meters. The larger the surface, the more efficient it produces.

Locomotive boiler Definition:

- A locomotive boiler is a multi-tubular boiler and it has a horizontal drum axis. The circulation in the locomotive boiler is natural, also it is a medium pressure boiler, the draft is artificial, forced circulation, mobile, solid fuel fired fire tube boiler.
- It is an external fire boiler and it has a large fire space. The locomotive boiler consists of a horizontal shell and this shell consists of a number of fire tubes.
- The fuel is burned here and the flue gases which pass through the shell and on the other side there is a stake locomotive boiler. It does not have the chimney and they have stacks on the other side and flue gases are removed through the stake.



Locomotive Boiler Main Parts or Construction:

Locomotive Boiler Consists of following Main Parts:

- Tubes
- Grates
- Ash pans
- Draft
- Diaphragm
- Stack
- Draft Pipe
- Smoke Box
- Safety Valve
- Brick Arches
- Water Level Indicator

Flues or Tubes:

A large part of the locomotive boiler is composed of flues or tubes. The flues give to the boiler the largest part of its heating surface.tubes are made up of carbon molybdenum steel, 18-8 Cr Ni steel.

It is the flues that largely affect the life of the boiler and that why it is the life of the locomotive, for this reason, it is quite necessary to properly install and maintain them.

Grates:

The grate is made up of a set of parallel bars at the bottom of the fire-box, which hold the fuel. it is made of cast iron and constructed in sections of 3-4 bars each.

They are supported at their ends by resting upon a frame and are connected by rods to a lever which can be moved back and forth to rack the bars and shake ashes and cinders out of the fire.

Ash Pans:

Ash pans are suspended beneath the fire-box for the purpose of catching and carrying the ashes. Coal that may drop between the grate bars is also corrected by ash pan. They are made of sheet steel.

A longitudinal section of an ash pan commonly used in fire-boxes placed between the axles of the engine. It is provided at each end with a damper a hinged at the top and which may be opened and set in any desired position in order to regulate the flow of air to the fire.

It is quite important that the dampers should be in good condition in order that the admission of air to the file may be regulated. The total unobstructed air openings in the ash pan need not exceed the total tube area.

Draft:

Draft receives power for doing this work from the exhaust steam from the cylinders. The work which it performs consists of drawing air through the ash pan, grates, fire, fire door, and other openings, then continues its work by drawing the gases of combustion through the flues of the boiler into the front end.

Pressure must be maintained less than the atmosphere in the smoke-box. The difference in pressure between the atmosphere and the smoke-box is called a draft.

Diaphragm:

The diaphragm or deflector plate is an iron plate placed obliquely over a portion of the front end of the flues which deflects the flue gases downward before entering the stack, thus equalizing to a great extent the draft in the different flues.

This deflector plate may be adjusted to deflect the gases more or less as desired.

Stack:

The stack is one of the most important features of the front end. Many different forms and proportions of stacks have been employed but at the present time, only two general types are found in use to any great extent, namely, the straight and tapered stacks.

Draft Pipes:

The draft pipe employed to increase the draft and may be used singly or in multiple and raised or lowered as desired.

Smoke-Box and Front End Arrangement:

By the term front end is meant all that portion of the boiler beyond the front tube sheet and includes the cylindrical shell of the boiler and all the parts contained therein such as the steam or branch pipes, exhaust nozzle, netting, diaphragm, and draft

Safety Valves:

The universal practice at present is to use at least two safety valves of the pop type upon every locomotive boiler. On small locomotives where clearances will permit, the safety valves are placed in the dome cap.

On large locomotives where the available height of the dome is limited, the safety valves are usually placed on a separate turret. When limiting heights will not permit the use of turrets, the safety valves may be screwed directly into the roof of the boiler.

Brick Arches:

A brick arch is an arrangement placed in the fire-box to effect better combustion and to secure a more even distribution of the hot gases in their passage through the tubes.

A longitudinal section of the fire-box fitted with a brick arch and the method of action is very simple. It acts as a mixer of the products of combustion with the air and as a reflector of the radiant heat of the fire and the escaping gases.

It is maintained at a very high temperature and in this condition meets the air and gases as they come in contact with it and turns them back to the narrow opening. By this action, it maintains a sufficiently high temperature to burn with the smallest possible quantity of air.

Water level indicator:

The water level indicator is used to indicate the water level in the boiler and to maintain the water at a constant level because the production of steam majorly depends on the quantity of water.

In case the water level is below the normal level then water will be pumped from the reservoir by the operator.

Locomotive Boiler Working Principle:

The locomotive boiler consists of a cylindrical barrel with a rectangular box at one end and a smokebox at the other end. The coal is introduced through the fire hole in the great which is placed at the bottom of the firebox.

The hot gases which are generated due to burning of the coal are deflected by an arch of fire bricks, so that walls of the firebox may be heated properly.

The firebox is entirely surrounded by water except for the fire hole and the ash pit which is situated below the firebox which is fitted with dampers at its front and back ends.

The dampers control the flow of air to the great. Hot gases move from the firebox to smokebox through a series of fire tubes and through the chimney they are discharged into the atmosphere.

The fire tubes are placed inside the barrel. Some of these tubes are of larger diameter and the others of smaller diameter. Tubes of superheater are placed inside the fire tubes of larger diameter.

The heating value of the hot gases is transmitted into the water by heating the surface of the fire tubes. The steam generated is collected over the water surface. Dome-shaped chamber known as the steam dome is fitted on the upper part of the barrel, from where the stream flows through a steam pipe into the chamber.

It passes through the superheater tubes and returns to the superheated steam chamber from which it is led to the cylinder through the pipes, one to each cylinder.

The boiler itself is moving with a very high velocity with 60 kilometres per hour to 70 kilometres per hour.

So there is always a movement of air over this stake and if you use Bernoulli's theorem you can find a temperature drop and there is a substantial pressure drop between this grate. The movement of the fluid pressure drop takes place so the chimney is not required in such boilers.

Advantages of Locomotive Boiler:

The following **advantages of Locomotive Boiler** are:

- High steam capacity
- Low cost of construction
- Portability is easy.
- Low installation cost

- Compact in design.

The following **disadvantages of Locomotive Boiler** are:

- Chances of corrosion and scale formation
- Difficult to clean some parts
- Need bracing for large parts
- It cannot carry overloads that create overheating and damage.

Application of Locomotive Boiler:

The **locomotive boiler is used at a various place** some of their application are:

- Railways, Road rollers.
- Agricultural fields
- Saw-Mill plants and
- Stationery power services.

Babcock and Wilcox Boiler Definition:

Babcock and Wilcox Boiler is a horizontal type drum axis, stationary, high pressure, natural circulation, solid fuel-fired water tube boiler in which coal is burned to heat the water for changing the phase into steam and later that steam is used for power generation.

Now comes to the construction,

Babcock and Wilcox Boiler Parts or Construction:

Babcock and Wilcox Boiler consist of the following Parts or Construction:

- Drum
- Inspecting Door
- Fire door
- Grate
- Ash pit

- Baffle Plate
- Upper take header
- Down header
- Feed Valve
- Water level Indicator
- Pressure gauge
- Safety Valve
- anti Primimg Pipe
- Main Stop Valve
- Manhole
- Superheater
- Drum
- Mud collector
- A blow of valve
- Damper
- Chimney and more

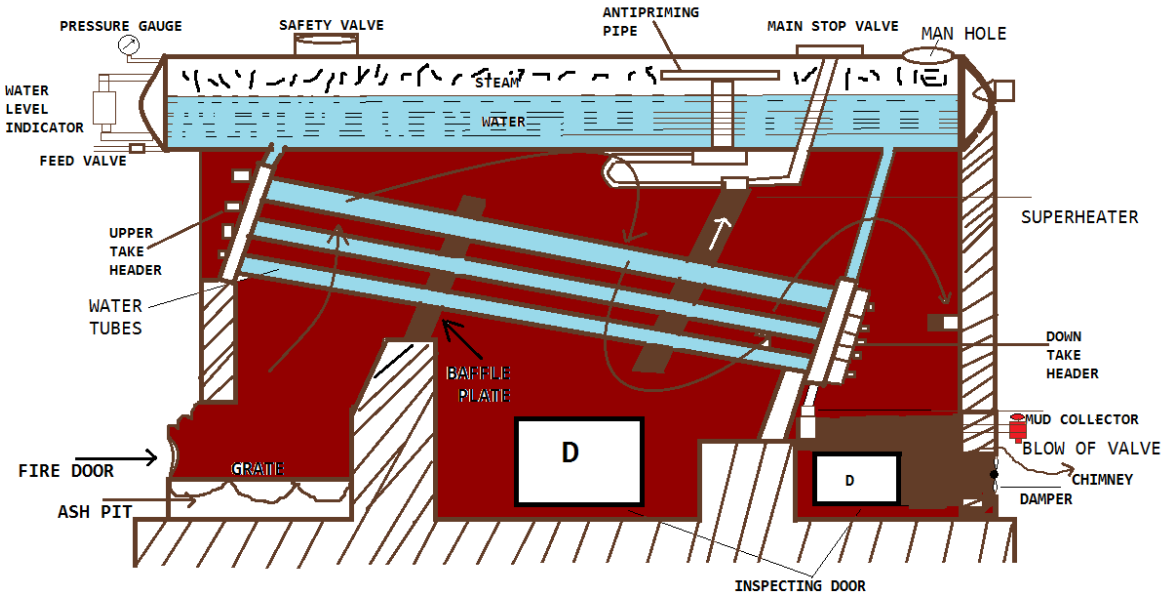


FIG: BABCOCK AND WILCOX BOILER

Drum:

It is a horizontal axis and it contains water and steam as you can see in the diagram.

Inspection Door:

It is provided for inspecting and cleaning the boiler.

Fire Door:

The Fire door is basically for providing the fire to start burning of fuel.

Grate:

It is basically known as the door for placing the fuel coal.

Ash Pit:

After the burning of fuel coal, the completely burned fuel which becomes ash now will come to the ash pit

It is located below the grate.

Baffle Plate:

The main function of the baffle plate is to divert the flue gases direction so that from one side to another side hot flue gases can easily go.

Upper and Down take header:

Upper take header connected to the drum from a very short pipe and down take header is connected to a long pipe.

The water comes from the drum through a long pipe to the down take header and it flows through the entire tube.

The steam formed will go up by a shorter tube (Upper take header).

Water level Indicator:

The Water level indicator is a measuring instrument. It measures the level of water and gives detail of the level of water.

Pressure gauge:

We use this instrument in several places. The works are to measure the pressure of the steam.

Safety Valve:

The safety valve is to prevent an increase in steam pressure.

Main stop valve:

The steam stop valve regulates the supply of steam outside to the boiler. First, from the boiler, the steam generated comes to an anti priming pipe here some parts of water particles will be removed and it completely remains dry steam then further this steam is used for needed place.

Man Hole:

Manhole is a very important part of the boiler. When parts or accessories get damaged or not working, then a boiler specialist engineer enters through the Man Hole and fixes the damaged parts.

Anti Primimg Pipe:

Anti priming pipe is provided in the boiler for extracting steam from the boiler is only dry saturated steam.

Mud Collector:

The impurities or contamination from the water will store here.

A blow of valve:

It is attached to the mud collector. Water contains some impurities like mud, sand, and salt. Due to the heating of water, these impurities deposited in the ground that creates a problem and it has to be removed.

So the Blow-off-valve is used for removing impurities from the mud collector.

Damper:

A Damper is here for removing the smokes from the boiler and it sends to the chimney.

Chimney:

The smoke released by the damper now comes to the chimney. From here smokes are released to the atmosphere. After studying construction now let’s have an overview of how it works,

Babcock and Wilcox Boiler Working Principle:

The working is, first the fuel (coal) is placed at the grate, and water is supplied to the drum. Now from the fire door ‘fire’ is supplied, to start burning of coal. The start burning of coal produces hot gases.

Water from the long pipe as you can see in the diagram (down take a header) is coming and impurities parts get into the mud collector and water is moving to the tubes.

The temperature of the flue gases increasing so water starts gets heating.

Here the baffle plate is placed to divert the flue gases direction to another side for an equal amount of flue gases for heating.

The water starts evaporating in the tube as increases the temperatures.

The steam formed will go up by short pipes to the drum and to the superheater to form completely dry steam and during these process circulation of water is on. The smoke released by the damper to the chimney and to the atmosphere.

Babcock and Wilcox Boiler Advantages:

The following advantages of the Babcock and Wilcox Boiler are:

- The overall efficiency of the Babcock and Wilcox boiler is high.
- This boiler occupies less space.
- The tubes used in the boiler can be exchanged easily when it gets damaged.
- A steam generation or production is high nearly around plus 20 tonnes per hour.
- The draught loss is limited.
- The transportation of the boiler is easy.

Babcock and Wilcox Boiler Disadvantages:

The following disadvantages of Babcock and Wilcox Boiler are:

- This boiler is not suitable for salty, Impure, and Sedemetry water. If these go to the boiler the overheating problem can occur that's why water treatment is necessary before inserting water for the working process.
- The maintenance and Installation cost is more.

Babcock and Wilcox Boiler Application:

This boiler is used for power plants for the generation of electricity and some other industries too.

Babcock and Wilcox Boiler Sailent Features:

The following Silent features of this Boiler are:

- Babcock and Wilcox Boiler, Overall efficiency is high compare to other fire tube boiler.
- The tubes which get defective can be replaced easily.
- The draught loss is minimum.
- The water tubes are kept inclined to 15 degrees for the circulation of water.
- The operating pressure and steam generation are high compare to another boiler.

Lamont Boiler Definition:

Lamont boiler is a type of water tube boiler in which steams are generated from water by the burning of fuel which is coal and later that steams are used for the generation of electricity and so on. It is a high-pressure boiler.

Lamont Boiler Parts or Construction:

A **Lamont boiler consists of the following parts or construction** as you can see in the diagram below:

- **Feed Pump**
- **Grate**
- **Economizer**

- **Circulating Pump**
- **Distributing header**
- **Radiant Evaporator**
- **Convective Evaporator**
- **Steam separating drum**
- **Superheater**
- **Blower**
- **Air preheater** and
- **Combustion chamber**

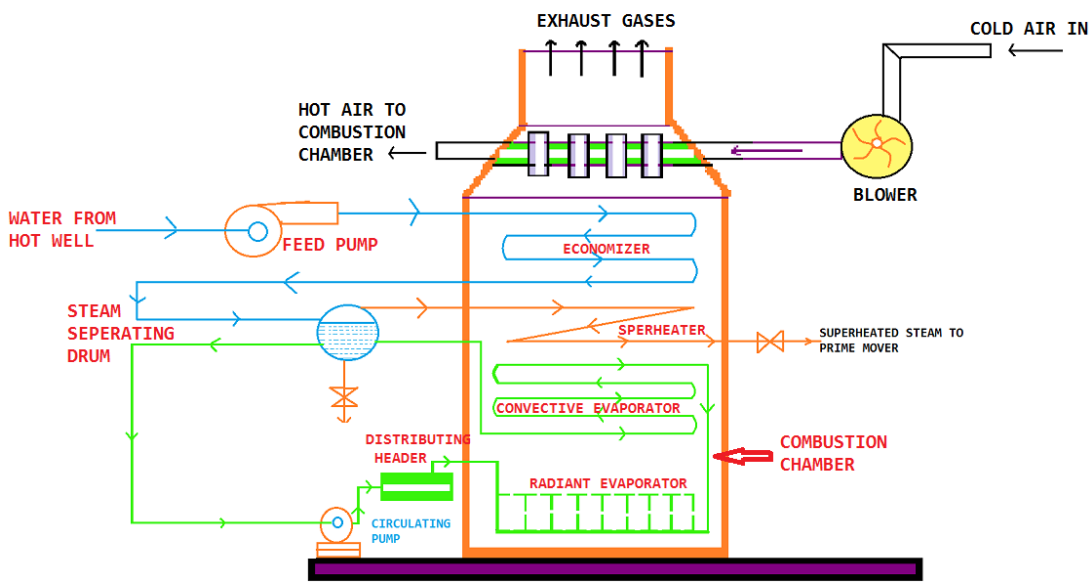


FIG: LAMONT BOILER

Feed Pump:

Feed Pump is one of the important types of pump which is used at the boiler for supplying the feed water. It increases the pressure so that the feed water can enter into the boiler.

Grate:

The grate is located bottom side of the furnace. At the grate, we supply coal for burning at the combustion chamber.

Economizer:

Economizer is a device for increasing boiler efficiency. So the work of the economizer is to preheat the water by the use of remaining heat flue gases of the combustion chamber.

In the repetition, the feed water is first entered into the economizer before entering into the boiler because it will preheat the water that is the major advantage of it.

Circulating Pump:

A circulating pump is a centrifugal pump and it is driven by the turbine. The turbine takes steam from the boiler. It takes water from the steam separating the drum and sends it to the Distributing header to the radiant evaporator section.

Distributing header:

It is a component having an inlet at one end only and more than one outlet end.

Radiant Evaporator:

The water starts changing phase from liquid to saturated liquid and to the saturated steam. The heat receives here from the burning of fuel that is coal.

Convective Evaporator:

Here the complete saturation of water into the saturated steam process takes place.

Steam separating drum:

The main work of the steam separating drum is to separate the steam and water. Further, the steam is sent to the superheater and water is sent to the circulating pump.

Superheater:

A name itself indicates “superheater” which means more heat will supply to the steam so that there should not be any tiny liquid particles and it will fully convert into the steam and for rotating the turbine blades.

Here we can say superheater increases the temperature of the steam.

Blower:

Here it is an external component that receives the air and sends it to the air preheater.

Air preheater:

Air Preheater also helps to increase boiler efficiency. Here the air is preheating and then preheating air sends to the combustion chamber.

Combustion chamber:

In the combustion chamber, the coal is burned and produces hot flue gases that help to heat the water.

Lamont Boiler Working Principle Diagram:

Lamont Boiler works on the principle of forced circulation of water in the boiler with the help of a centrifugal pump.

This is a high-pressure boiler and works above 170 bar and temperature around 773k.

we will be understanding here in pointwise in depth so let’s start,

The water is first supplied to the feed pump from a large water tank. From the feed pump, the water is entered into the economizer.

As we know it is a boiler efficiency increasing device so here if heat is available then the water will get heated before entering into the combustion chamber.

From the economizer, the water comes to a steam separator device where the separation of steam and water takes place.

The water is circulated by the centrifugal pump from the separator drum. Here distributor header is kept for controlling the water level entry into the boiler.

Now water reaches into the combustion chamber where flue gases surround them and it starts getting heating.

Now the water comes in the phase of radiant evaporator here major amount of water is converted into steam and then it sends to the convective heat transfer.

Here the remaining water is further converted into saturated steam. Now, this saturated steam is further sent to the separator drum for separating steam.

After separating steam now further sent to the superheater to superheat the saturated steam and make use of it further for the generation of electricity and more.

Lamont Boiler Advantages:

- In terms of the design of the boiler are very simple and understandable.
- The drum size is small.
- It is easy to start. There is no such problem with starting.
- Steam generation capacity is very good and it is 50 tonnes per hour.
- The transfer rate of heat is also good and the major drawback is bubble formation.

Lamont Boiler Disadvantages:

The **major disadvantages of the Lamont boiler are bubble formation at the surface of the tubes that reduces the transfer rate of heat and causes low steam generation**.

Lamont Boiler Application:

A Lamont Boiler has the following application or uses:

- Lamont Boiler is mostly used in Power plants for producing steam which helps to run a turbine for the generation of electricity.
- It is also used in other industries like Sugar Mill, Chemical industry, and more.

Benson Boiler Definition:

Benson Boiler is a high-pressure super critical water type boiler in which no bubble is formed because the water is compressed at super critical pressure and it directly converts into steam.

In other words, The formation of a bubble does not take place because of supercritical pressure, and at this pressure, the density of water and steam becomes the same.

As the water which is entered into the boiler is being compressed to supercritical pressure. Therefore the latent heat of water is reduced to zero. So the water directly changes into steam without the formation of bubbles.

Benson Boiler Parts or Construction:

It consists of following Main Parts:

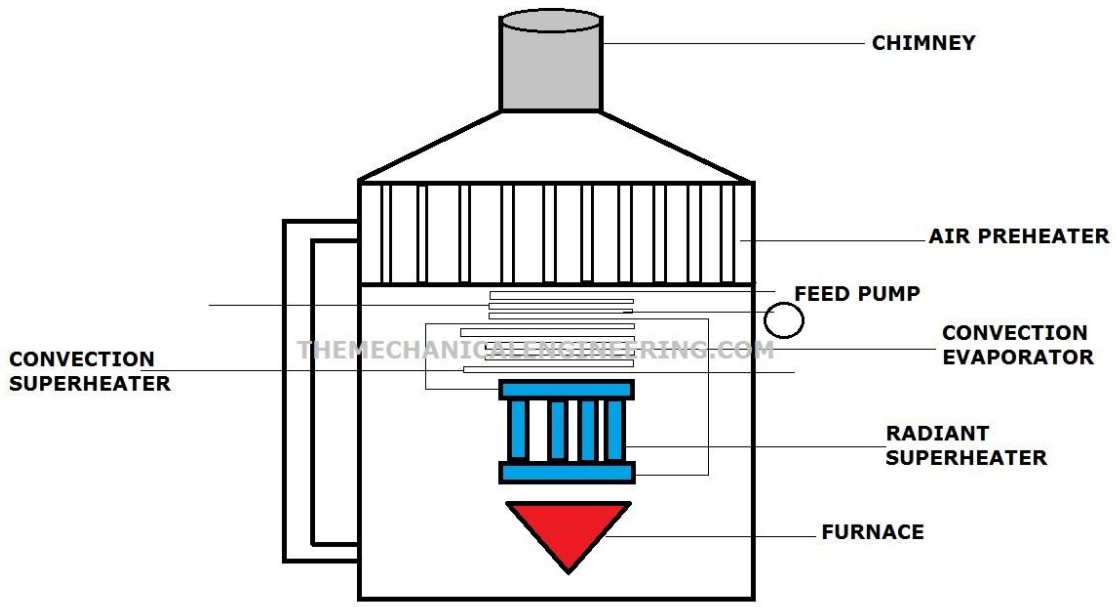
- Chamber
- Water Feed Pump
- Blower
- Furnace
- Economizer
- Air Preheater
- Radiant Superheater
- Convection Evaporator
- Convecting Superheater
- Throttle Valve
- Chimney

Benson Chamber:

It is a outer parts of this boiler where all the other parts of this boiler are kept.

Water Feed Pump:

The work of this feed pump is to supply the water from the hot well to inside the boiler at a supercritical pressure of 225 bars.



LINE DIAGRAM OF BENSON BOILER

Blower:

As in general we know that the work of a blower is to blow the air in a particular direction at a very high speed. The blower is connected with the air preheater and its working is the same. The blower blows air into the air preheater and the air preheater will heat the air.

Furnace:

The furnace of any boiler where fuel is burnt and here it has a maximum temperature. The flue gases are passed to the tube and other places from here only.

Economizer:

Economizer's contribution is to increase the efficiency of any boiler. Here The water which is entered through the feed pump now it passes through the economizer and This heats the water at a certain temperature with the help of combustion gases inside it.

Air Preheater:

The name itself indicates "preheat" that means the air is being heated before entering to the boiler so we can say it also helps to increase the overall efficiency of the boiler.

Radiant Superheater:

Again radiant superheater helps to heat the water with radiation produced by burnt fuel, therefore, the temperature of the water rises to supercritical temperature and It is connected to the convective superheater and that is finally connected to the turbine.

Convection Evaporator:

In this evaporator, the water evaporates completely. In other words, the Convection evaporator evaporates the superheated water and converts them into steam. The convection process of heat transfer is used to transfer heat from the flue gases to water.

Convection Super heater:

The conviction super heater work is to superheats the steam to the desired temperature which is at 650 degree Celsius.

Chimney:

The burnt gases which is now became smokes will be releases by the chimney to the atmosphere.

Throttle valve:

It is types of valve which work is to store and control the flowing passages of steam.

Now we will study Benson Boiler working in detail,

Benson Boiler Working Principle:

The water feed pump sends the water into the economizer (here it increases the pressure of the water to the supercritical pressure). Now In the economizer, the water is preheated using combustion gases that help to increase the efficiency of the boiler.

Now the water from From economizer which is already heated at a certain temperature passes to the radiant superheater for further heating.

In the radiant super heater, the water is heated using the radiant heat transfer method.

Here the heat is received from the combustion chamber. Now the water will partially convert into steam and some amount also remains in the liquid form

which is not converted into steam completely. So from here, it is further sent to the superheater for converting completely into steam.

In the convective superheater, The remaining water or partial steam is fully converted into vapor or steam and this superheated steam will go to the prime mover for the rotation of the turbine. Further, it is being used for power generation.

Benson Boiler Advantages:

- The Benson boiler cost is cheaper as there is no presence of a steam drum.
- It produces higher power pressure, so there is no pressure limit to attain supercritical pressure.
- The attaining of high pressure avoids the bubble collection in the tubes and increases the maximum heat rate. The presence of bubbles in the boiler causes lower efficiency power of the boiler.
- It’s lightweight than any other boilers.
- The boiler is efficient in running, starts in 15mins due to welded joints.
- Boiler transportation is simple and easy.
- The boiler is efficient and achieves a thermal efficiency of up to 90%.
- The boiler consists of small tubes with lesser diameters. It helps in creating negligible situations prone to explosion hazard risks.

Benson Boiler Disadvantages:

The **disadvantages of Benson Boiler** is:

- An overheating of the tubes can cause insufficient water supply.
- Since the storage capacity is small, it requires close coordination between steam, feedwater, and feed inlet.
- The evaporation process gets accompanied by the salts and solid in tubes if the water present is impure. It can cause severe damage and blockage to the boiler tubes.
- You may face some problems and difficulties controlling the boiler for variable loads.

- Its movements require continuous inspection in every duration to prevent an explosion. It happens due to the presence of supercritical pressure.

Benson Boiler Application:

The main application of Benson Boiler is:

- The high-pressure supercritical boiler is used in various industry, to produce steam and generate electrical power.
- The boiler's average optimum pressure is 250 bar with a temperature of 50 degrees celsius and a capacity to produce about 135 tons/ hour.

Benson Boiler Features:

- The weight of this boiler is comparatively low than the other boilers. It makes it cheap and reduces production costs.
- The boiler consists of a lack of drums. It makes the parts transfer easy.
- They are transported easily without the need to preassemble them.